4.2. WHY ARE STOCKS A BAD INFLATION HEDGE?

There are a number of reasons why stocks do badly in periods of high inflation. I will examine two rational stories in the context of Ang and Ulrich (2012), where I develop a model where equities are influenced by the same factors driving bond prices and an equity-specific cash flow factor.

First, high inflation reduces future firm profitability. Inflation is negatively associated with real production; see, for example, Fama (1981). Rising inflation reduces profit margins because, while firms can pass through cost increases to consumers continuously, they can only do so in stages. These are called "*menu costs*" and, empirically, price rigidity is pervasive, as Nakamura and Steinsson (2008) show. This is a *cash flow* effect.

Second, there is also a *discount rate* effect. Times of high inflation are bad times when risk is high. This causes expected returns to be high in a factor story. Even in the consumption-only world of Mehra and Prescott, the historical record has seen high inflation episodes occurring with recessions when growth is slow (stagflation). Thus, when inflation is high, expected returns on equity increase, and this cuts equity prices. Thus, equity prices tend to fall with large inflation increases, resulting in a low correlation between *realized* inflation and *realized* equity returns. If this were always true, the correlation between inflation and equity should be −1. It is not because there are other factors driving equity prices that are not perfectly correlated with inflation.

The behavioral explanation of equities' shortcoming as an inflation hedge was proposed by Modigliani and Cohn (1979). According to this account, investors suffer from *money illusion*, and they discount real dividends using nominal discount rates instead of using real discount rates. Thus when inflation is high, the market's irrational expectation causes the market to be undervalued relative to the fundamental value. Thus, in times of high inflation, there are realized low returns on the market leading to low correlations of inflation with stock returns.

Under both the rational and behavioral explanations, equity has low returns during high inflation periods and is a bad inflation hedge. Inflation is a risk factor, but inflation causes equities to perform badly, not well. Consequently, any investor forecasting high inflation in the future relative to what the average investor believes is advised to lower holdings of equities.

5. Predicting Equity Risk Premiums

Active investors spend a lot of effort trying to forecast equity returns. But the consensus view in financial theory is that, while equity risk premiums vary over time, movements are hard to predict. Thus, I advise most investors not to time the market, and this is behind my advice in chapter 4 to rebalance back to constant weights or exposures.